ESP32 Fleet Management — Component Reference

Tetradapt | Cloudflare + DigitalOcean | $24/mo

Cloudflare — EXTERNAL — FREE (EXISTING ACCOUNT)

COMPONENT	PORT / PROTOCOL	PURPOSE
Cloudflare DNS	–	Resolves `fleet.tetradapt.com` to Droplet IP. Orange cloud proxy enabled.
Cloudflare SSL	`:443 HTTPS`	Terminates SSL at edge. Full (Strict) mode — encrypts edge-to-origin. Free universal cert.
Cloudflare CDN	–	Caches `/firmware/*.bin` (1h edge TTL) and `/dashboard/*` (24h) at 300+ global PoPs. ESP32 pulls firmware from nearest edge.
Cloudflare DDoS	–	Automatic DDoS mitigation on all proxied traffic. Always-on, no configuration needed.
Cloudflare WAF	–	Web Application Firewall — blocks malicious requests, SQL injection, XSS before reaching origin.
Cloudflare Rate Limiting	–	Throttles `/api/broadcast` to 10 req/min. Prevents broadcast flood abuse.

DigitalOcean Free Services — FREE TIER

COMPONENT	PORT / PROTOCOL	PURPOSE
DO Cloud Firewall	–	Stateful firewall rules: port 443 (Cloudflare IPs only), 8883 (home VPN IPs), 51820 (WireGuard). Blocks all other inbound.
DO VPC	`10.0.0.0/16`	Private network in SFO3 region. Isolates Droplet from other tenants. Free internal traffic.
DO Monitoring	–	Built-in system metrics — CPU, RAM, disk, bandwidth. Email and Slack alerts on threshold breach.
DO Container Registry	–	Stores `fleet-server:latest` Docker image. Free tier: 1 repo, 500MB storage.

Application Layer — Docker Compose on Droplet ($24/mo) — 2 vCPU / 4GB RAM / 80GB SSD

COMPONENT	PORT / PROTOCOL	PURPOSE
Nginx	`:80 HTTP`	Reverse proxy — routes `/api/*` to FastAPI, serves `/dashboard/*` static files, serves `/firmware/*.bin` for CDN origin, upgrades `/ws/*` to WebSocket. SSL handled by Cloudflare.
FastAPI	`:8000`	Fleet management REST API — CRUD for devices, homes, communities, firmware. Triggers OTA rollouts, composes/sends broadcast messages, queries logs and audit trail. WebSocket for live fleet status + OTA progress. Broadcast uses FCM push to mobile apps.
APScheduler	`(in-process)`	In-process async task scheduler inside FastAPI. Replaces Celery + Redis. Runs: `ota_rollout()` canary/staged, `send_broadcast()` community fan-out, `upload_firmware()` validation, `scheduled_broadcast()` timed delivery to mobile app subscribers.
React Dashboard	`(static files)`	Web UI served by Nginx, cached by Cloudflare CDN. Pages: Fleet Overview, Device Detail, OTA Manager, Broadcast Center (compose/schedule/delivery report), Communities, Log Viewer, Security/Audit. Real-time updates via WebSocket.

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Data Layer — Docker Compose (same Droplet) FREE — SELF-HOSTED

COMPONENT	PORT / PROTOCOL	PURPOSE
PostgreSQL 16	:5432 TCP	Primary relational database. Tables: devices, homes, communities, home_community (many-to-many), broadcasts, broadcast_acks, firmware_versions, ota_events, audit_log, users. Docker volume: /pgdata.
Mosquitto MQTT	:8883 TLS	Message broker with TLS + ACL authentication. Topics: fleet/{device}/heartbeat (30s interval), fleet/{device}/register (on boot), fleet/broadcast/community/{group_id} (server → homes fan-out), fleet/broadcast/home/{home_id} (direct), fleet/{home_id}/broadcast/ack (delivery receipt), fleet/{device}/ota/cmd + status.
WireGuard VPN	:51820 UDP	VPN server — encrypted tunnel from each patient home to Droplet. Bypasses Cloudflare (direct IP). Carries MQTT traffic and syslog UDP. Each home gets a unique peer config with pre-shared key.

Monitoring Layer — Docker Compose (same Droplet) FREE — OPEN SOURCE

COMPONENT	PORT / PROTOCOL	PURPOSE
Grafana OSS	:3000 HTTP	Monitoring and visualization dashboard. Datasources: Loki (logs), PostgreSQL (fleet metrics). Dashboards: device health heatmap, fleet uptime %, OTA rollout tracker, broadcast delivery rates, RSSI signal map. Alerting via email/Slack.
Loki	:3100 HTTP	Log aggregation engine. Indexes and stores device syslog data from Rsyslog. Queryable via LogQL in Grafana. Low resource — indexes labels only, stores log lines compressed.
Rsyslog	:514 UDP	Syslog collector. Receives ESP32 device logs over UDP via WireGuard VPN. Parses, tags by device name, forwards structured logs to Loki for indexing and search.

Local Storage — Droplet 80GB SSD INCLUDED WITH DROPLET

PATH	FORMAT	PURPOSE
/data/firmware/*.bin	.bin (1–2MB each)	ESPHome compiled firmware binaries. Served by Nginx at /firmware/, cached globally by Cloudflare CDN. ESP32 devices pull from nearest CF edge during OTA.
(no broadcast storage)	—	Broadcasts are MQTT pub/sub only — messages go directly to SmartHome mobile apps via MQTT topics. No files stored on server.
/data/backups/*.sql.gz	.sql.gz (~50MB)	Daily PostgreSQL dumps via pg_dump cron (2am). 7-day retention, auto-cleanup of older files. Full disaster recovery via DO Droplet snapshots ($0.06/GB).
/pgdata/	Docker volume	PostgreSQL persistent data directory. Survives container restarts and redeployments.

Background Services — Inside FastAPI Process ALWAYS RUNNING

SERVICE	TRIGGER	PURPOSE
Heartbeat Monitor	Every 30s	Processes MQTT heartbeat messages (uptime, free heap, RSSI). Updates device status in PostgreSQL. Marks device dead after 90s silence. Triggers Alert Service.
Device Registrar	On MQTT register	Auto-creates new devices in PostgreSQL when ESP32 boots and publishes MAC/version/role. Assigns device to home and community group.
Broadcast Scheduler	Cron + on-demand	Fires scheduled broadcasts at configured times. Enforces Do-Not-Disturb hours per home. Retries undelivered messages for homes that were offline.

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Patient Home — Edge (per home) NRH | KAISER | ST JUDE | SUTTER HEALTH

COMPONENT	PORT / PROTOCOL	PURPOSE
Home Assistant	:8123 HTTP	Smart home hub. Runs automations, manages ESP32 devices via ESPHome API. Local device control only — broadcasts go to mobile app, not HA.
WireGuard Client	:51820 UDP	VPN tunnel back to fleet server Droplet. Encrypts all MQTT and syslog traffic. Auto-reconnects on network changes.
SmartHome App	MQTT :1883	iOS/Android app subscribes to `fleet/broadcast/community/{group}` MQTT topics. Shows push notifications. Publishes delivery ACKs back to fleet server. Separate from HA.

ESP32 Devices — ESPHome Firmware (per device) N DEVICES PER HOME

COMPONENT	INTERVAL / TRIGGER	PURPOSE
ESPHome Firmware	Always running	YAML-configured firmware. Handles WiFi, MQTT, OTA, sensors, heartbeat. No C++ coding — declarative config, version-controlled in Git.
MQTT Heartbeat	Every 30s	Publishes to `fleet/{device}/heartbeat` — JSON payload: uptime (seconds), free heap (bytes), WiFi RSSI (dBm).
MQTT Register	On boot	Publishes to `fleet/{device}/register` — MAC address, firmware version, device role.
OTA Client	On MQTT command	Receives OTA command via MQTT. Downloads `.bin` from Cloudflare CDN URL. Validates checksum, flashes, reboots. ACKs success/failure.
Syslog Client	On log event	Sends device logs via UDP syslog to fleet server Rsyslog (:514) through WireGuard VPN.
Sensors	Varies	Fall detection, emergency button, motion sensor, door sensor — varies per home and patient needs. Reports to Home Assistant via ESPHome API.

Authentication & Authorization OAuth2 / JWT

ROLE	ACCESS LEVEL	PERMISSIONS
admin	Full access	All API routes. OTA rollouts, broadcast compose/send, device/home/community CRUD, firmware upload, delete operations, audit trail, user management.
operator	Operations	OTA rollouts, broadcast compose/send, device and community management. No delete operations, no user management.
viewer	Read-only	Dashboard viewing, fleet status, device health, broadcast delivery reports, log search. No mutations.